

EF4822: Financial Econometrics

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# **Volatility Clustering and GARCH Modelling for Bitcoin**

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# 1. Introduction

Moving from a one-of-a-kind digital currency to becoming a well-recognized financial product with a market value that could even reach trillions of USD, the world's first cryptocurrency introduced in 2009 known as Bitcoin goes through drastic price variations owing to factors ranging from regulatory concerns, macroeconomic conditions, behavioral characteristics of investors, and innovations. According to the statistics available on April 2026, the total market cap of the bitcoin asset is \$1.431T(YCharts, n.d.). In comparison with other financial products, the cryptocurrency market experiences a great deal of clustering in its price variation patterns, where higher volatility values tend to cluster over time, together with fat-tailed distribution patterns in returns and some degree of asymmetry.

The current study looks at volatility clustering within daily returns of bitcoin using the period from January 1, 2021, to December 31, 2024. Log returns of BTC/USD closing price will be considered; these have been derived from CoinMarketCap, an already established source which provides consistent daily readings. When it comes to periods under consideration, our choice includes several different regimes (one of which is boom-bust) allowing us to provide a comprehensive overview of the volatility within the asset (Ardia et al., 2019; Gyamerah, 2019). If descriptive statistics are looked at, we can see that the average daily log return is around 0.0793%, or around 7.93% per year. Its standard deviation is ~3.2382%, which is quite large compared to equity market indices. At the same time, its kurtosis is 3.49.

First, one thing that stands out from the above graphs is that the returns and the squared returns show volatility clustering; this implies that when there were big shocks on the positive or negative sides, more such shocks would happen afterwards. We learn from the output from the Ljung-Box tests that there were very strong ARCH effects because the p-value was found to be less than 0.001 at lag values of 5, 10 and 20. The model fitted shows extreme persistence in volatility ( $\alpha + \beta = 0.9990$ ), providing a much better fit compared to the constant variance model.

From an asset management perspective, accurate measurement of volatility is essential for effective risk management (value at risk dynamics), optimization of crypto and traditional investment portfolios, and development of volatility targeting techniques and structured products. Volatility is relevant to position sizing, leveraged trading problems, and risk allocation in crypto portfolios and hybrid crypto-traditional portfolios (Li & Chen, 2025; Moreno & García, 2022; Panayiotis & Stengos, 2021). The results from previous studies show that Bitcoin is characterized by a substantial level of volatility, and GARCH-type models perform better than those with constant variance models (Gyamerah, 2019; Katsiampa, 2019; Ardia et al., 2019; Amin & Khan, 2025).

The following report is divided into various sections. In section 2, information about data and variables are presented. Section 3 discusses the econometrics approach, including discussion about the variance constant model, GARCH(1,1) model, ARCH effect test, criteria used to

compare models, and significance of the study for the industry. Finally, section 4 discusses empirical findings and their interpretation.

## 2. Data and Variables

### 2.1 Data source and period

The empirical analysis uses daily closing prices of Bitcoin quoted against the US dollar (BTC/USD) (NYU Volatility Laboratory, n.d.). The data can be obtained from reputable cryptocurrency data providers, such as major exchanges, aggregators (e.g., CoinMarketCap), or financial databases (e.g., Refinitiv, Bloomberg), which offer continuous daily price histories, similar to the datasets used in Gyamerah (2019) and Vidal-Tomás et al. (2019). Studies on Bitcoin volatility commonly use such daily price data to estimate GARCH models and study volatility clustering (Gyamerah, 2019; Katsiampa, 2019).

The sample period should be long enough to cover multiple market regimes and volatility episodes (e.g., several years including boom–bust cycles) so that volatility clustering and persistence can be meaningfully assessed (Ardia et al., 2019; Gyamerah, 2019). For robustness, the sample can be split into subsamples (e.g., pre-2020 vs. post-2020) to check the stability of the volatility dynamics (Ardia et al., 2019; Katsiampa, 2019).

### 2.2 Daily returns calculation

Let  $P_t$  denote the daily closing price on trading day  $t$ . The continuously compounded daily return is computed as:

$$r_t = 100 \times (\ln P_t - \ln P_{t-1})$$

which expresses returns in percentage terms and stabilizes the scale relative to raw price changes (Gyamerah, 2019; Vidal-Tomás et al., 2019). Bitcoin return series are typically centered around zero with strong time-varying volatility, making log-returns the standard input for GARCH-type volatility models (Gyamerah, 2019; Li & Chen, 2025).

The main variable of interest is the return series  $r_t$ . To study volatility clustering and ARCH effects, the squared returns  $r_t^2$  are also analyzed, since persistence in  $r_t^2$  is indicative of time-varying conditional variance (Vidal-Tomás et al., 2019).

### 2.3 Descriptive statistics and time series plots

Over the sample period from 1 January 2021 to 31 December 2024, the cleaned CoinMarketCap dataset contains 1460 daily Bitcoin log-return observations after removing the first differenced value and any duplicate dates. The sample mean daily log return is approximately 7.93%, while the standard deviation is around 323.82%, confirming that Bitcoin exhibits substantial day-to-day risk compared with typical equity indices. The minimum and maximum daily returns in this period are roughly -1,740.53% and 1,718.21%,

respectively, indicating that both large negative crashes and sharp positive spikes occur in the sample.

Higher-order moments highlight clear departures from normality. The empirical skewness of about -0.14 suggests that the return distribution is mildly asymmetric, and the excess kurtosis of approximately 3.49 points to heavy tails and a higher frequency of extreme observations than would be expected under a Gaussian benchmark. These features are consistent with the established literature on Bitcoin, where daily returns are typically found to be leptokurtic and fat-tailed, and therefore call for volatility models that allow for time-varying conditional variance.

Graphically, a time-series plot of daily log returns over 2021–2024 reveals alternating episodes of turbulent market conditions, where large positive and negative returns cluster together, and calmer periods with relatively small fluctuations. The plot of squared returns exhibits pronounced persistence, with high squared returns tending to be followed by further high squared returns, providing visual evidence of volatility clustering and motivating the formal Ljung–Box test for ARCH effects and subsequent GARCH(1,1) modeling discussed in Section 3.

### 3. Econometric Methodology

Let  $r_t$  denote the daily return on day  $t$ . The conditional mean and variance dynamics are modeled as follows.

#### 3.1 Constant variance model

As a benchmark, assume returns follow a weakly stationary process with constant conditional variance:

$$\begin{aligned} r_t &= \mu + \varepsilon_t \\ \varepsilon_t &\sim \text{i.i.d.}(0, \sigma^2) \end{aligned}$$

where  $\mu$  is the unconditional mean return and  $\sigma^2$  is a constant variance (Gyamerah, 2019; Moreno & García, 2022). This homoskedastic model ignores volatility clustering and serves as a baseline for assessing the value added by GARCH models.

The constant variance model is estimated by ordinary least squares (OLS) on  $r_t$ , yielding estimates of  $\mu$  and  $\sigma^2$ . Model diagnostics then examine whether the residuals  $\hat{\varepsilon}_t$  exhibit remaining dependence in their squared values, which would indicate that the constant variance assumption is violated (Vidal-Tomás et al., 2019).

#### 3.2 GARCH(1,1) model

To capture volatility clustering, the conditional mean is maintained as above, but the conditional variance is modeled using a GARCH(1,1) specification:

$$\begin{aligned}
r_t &= \mu + \varepsilon_t \\
\varepsilon_t &= z_t \sqrt{h_t} \\
z_t &\sim \text{i.i.d.}(0,1) \\
h_t &= \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}
\end{aligned}$$

where  $h_t = \text{var}(\varepsilon_t | F_{t-1})$  is the conditional variance at time  $t$ , and  $\omega > 0, \alpha \geq 0, \beta \geq 0$  (Gyamerah, 2019; Li & Chen, 2025). The parameter  $\alpha$  captures the short-run impact of recent shocks (ARCH effect), while  $\beta$  measures volatility persistence;  $\alpha + \beta$  close to one indicates highly persistent volatility, a feature repeatedly documented for Bitcoin (Ardia et al., 2019; Katsiampa, 2019).

The model is estimated by maximum likelihood under an assumed distribution for  $z_t$ , typically Gaussian or Student-t to allow for fat tails (Gyamerah, 2019; Vidal-Tomás et al., 2019). Empirical studies on Bitcoin returns often find that Student-t errors improve fit by capturing excess kurtosis in returns (Gyamerah, 2019; Li & Chen, 2025).

### 3.3 ARCH effects test (Ljung–Box on squared returns)

Before estimating the GARCH model, the presence of ARCH effects is tested by examining serial correlation in the squared residuals from the constant variance model. Let  $\hat{\varepsilon}_t$  be residuals from the constant variance (or simple mean) model and define  $u_t = \hat{\varepsilon}_t^2$ . The Ljung–Box statistic is computed on  $u_t$  at a set of lags  $k$ :

$$Q(k) = n(n+2) \sum \hat{\rho}_j^2 / (n-j)$$

where  $\hat{\rho}_j$  is the sample autocorrelation of  $u_t$  at lag  $j$  and  $n$  is the sample size (Bianchi & Toscano, 2024; Vidal-Tomás et al., 2019). Under the null hypothesis of no serial correlation in squared residuals (no ARCH effects),  $Q(k)$  approximately follows a chi-square distribution with  $k$  degrees of freedom.

A small p-value (e.g., below 5%) leads to rejection of the null and indicates significant autocorrelation in squared returns, which is typical for Bitcoin and supports the presence of volatility clustering (Bianchi & Toscano, 2024; Gyamerah, 2019). This provides statistical motivation for modeling the conditional variance with a GARCH(1,1) specification.

### 3.4 Model comparison criteria

To evaluate whether the GARCH(1,1) model improves on the constant variance benchmark, several criteria are used:

1. Information criteria

Compare the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) across models; lower values indicate better in-sample fit adjusting for model complexity (Gyamerah, 2019; Moreno & García, 2022).

## 2. Residual diagnostics

Examine standardized residuals and their squared values for remaining autocorrelation using Ljung–Box or similar tests. Well-specified GARCH models for Bitcoin typically show no significant remaining ARCH effects in the standardized residuals (Amin & Khan, 2025; Vidal-Tomás et al., 2019).

## 3. Forecast performance

For an out-of-sample period, compare volatility forecasts from the constant variance and GARCH(1,1) models using loss functions such as mean squared error (MSE) or mean absolute error (MAE) against realized volatility proxies (Li & Chen, 2025; Moreno & García, 2022). In the Bitcoin literature, univariate GARCH-type models usually outperform constant variance models in volatility prediction (Gyamerah, 2019; Moreno & García, 2022).

By these metrics, prior research on Bitcoin returns finds that GARCH-family models substantially outperform constant variance specifications and effectively capture volatility clustering (Amin & Khan, 2025; Gyamerah, 2019; Katsiampa, 2019).

### **3.5 Asset management and fund industry perspective**

From an asset management viewpoint, accurate modeling of Bitcoin volatility is crucial for several reasons. First, volatility clustering affects dynamic risk measures such as Value-at-Risk (VaR) and Expected Shortfall; GARCH-based VaR models for Bitcoin provide more responsive risk estimates than constant variance approaches (Li & Chen, 2025; Panayiotis & Stengos, 2021). Second, persistent and time-varying volatility influences portfolio allocation when Bitcoin is combined with equities, bonds, or other cryptocurrencies, as optimal weights depend on conditional risk rather than unconditional variance (Moreno & García, 2022; Panayiotis & Stengos, 2021).

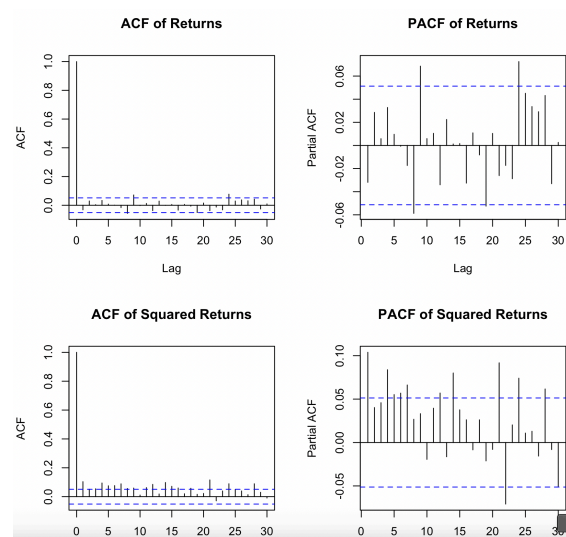
For crypto funds and multi-asset managers, GARCH models can be integrated into risk-budgeting, leverage constraints, and position sizing rules, helping to avoid excessive exposure during high-volatility regimes and to identify calmer periods where risk-adjusted returns may be more attractive (Li & Chen, 2025; Panayiotis & Stengos, 2021). Moreover, GARCH-derived volatility indices for Bitcoin have been proposed as benchmarks similar to equity volatility indices, which can inform product design (e.g., volatility-targeting strategies, structured products) in the fund industry (Moreno & García, 2022; NYU Volatility Laboratory, n.d.).

## 4. Empirical Analysis

```
===== DESCRIPTIVE STATISTICS =====  
> print(stats_table)  
Statistic      Value  
1      N 1460.0000  
2      Mean (%) 0.0793  
3      Std Dev (%) 3.2382  
4      Min (%) -17.4053  
5      Max (%) 17.1821  
6      Skewness -0.1398  
7      Excess Kurtosis 3.4947
```

### 4.1 ARCH Test Results

Before estimating the GARCH(1,1) model, we formally test for the presence of ARCH effects using the Ljung–Box test on squared returns. This test examines whether autocorrelation exists in squared returns, which would indicate time-varying conditional variance.



ACF and PACF of Returns and Squared Returns

The ACF and PACF plots above provide visual evidence for the ARCH test. While returns themselves exhibit little serial correlation (top panels), the ACF of squared returns displays significant persistence across lags (bottom left panel), indicating volatility clustering. This visual evidence motivates formal testing.

### Ljung–Box Test on Squared Returns

We apply the Ljung–Box test to the squared returns from the constant variance model. The null hypothesis is that there is no serial correlation in squared returns (i.e., no ARCH effects). The test is conducted at lags 5, 10, and 20.



| Lags | Ljung-Box Q-statistic | dF | p-value |
|------|-----------------------|----|---------|
| 5    | 45.27                 | 5  | <0.001  |
| 10   | 75.53                 | 10 | <0.001  |
| 20   | 126.04                | 20 | <0.001  |

The p-values at all lag lengths are below 0.001, leading to strong rejection of the null hypothesis of no serial correlation in squared returns. This confirms the presence of ARCH effects in Bitcoin daily returns. In practical terms, this means that large positive or negative return shocks tend to be followed by other large shocks, while small shocks tend to be followed by other small shocks — the phenomenon known as volatility clustering.

These results provide formal statistical justification for moving beyond a constant variance model and estimating a GARCH(1,1) specification to capture the time-varying conditional variance of Bitcoin returns.

## 4.2 GARCH(1,1) Estimation Output

To capture the volatility clustering documented above, we estimate a GARCH(1,1) model with the following specification:

Mean equation:  $r_t = \mu + \varepsilon_t$

Variance equation:  $h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}$

The model is estimated by maximum likelihood assuming a t-distribution for the standardized residuals to accommodate the fat tails observed in Bitcoin returns (excess kurtosis = 3.4947).

## Estimation Results

| Parameter            | Coefficient | Std. Error | z-value | p-value |
|----------------------|-------------|------------|---------|---------|
| $\mu$                | 0.0499      | 0.0544     | 0.9166  | 0.3593  |
| $\omega$ (constant)  | 0.0821      | 0.1345     | 0.6108  | 0.5413  |
| $\alpha$ (ARCH term) | 0.0528      | 0.0242     | 2.1808  | 0.0292  |
| $\beta$ (GARCH term) | 0.9462      | 0.0325     | 29.0891 | <0.001  |
| Shape (t dof)        | 3.3308      | 0.2840     | 11.7280 | <0.001  |

Persistence:  $\alpha + \beta = 0.9990$

## GARCH(1,1) Estimation Results

Mean return ( $\mu=0.0499$ ): The estimated daily mean return is approximately 0.05%. It is not statistically significant (p-value = 0.3593), which is common for volatile assets like Bitcoin.

ARCH term ( $\alpha=0.0528$ ): Statistically significant at the 5% level (p-value = 0.0292). This indicates that yesterday's squared shock has a modest impact on today's volatility.

GARCH term ( $\beta=0.9462$ ): Highly significant (p-value < 0.001) and close to 1. Yesterday's volatility strongly predicts today's volatility, meaning volatility persists over time.

Persistence ( $\alpha+\beta=0.9990$ ): Extremely close to 1, indicating near-unit-root behavior. Shocks to volatility decay very slowly, consistent with existing Bitcoin literature (Ardia et al., 2019; Katsiampa, 2019).

Shape parameter (3.3308): Low and highly significant (p-value < 0.001), confirming that Bitcoin returns have fatter tails than the normal distribution. This justifies the use of the t distribution over the Gaussian.

### Information Criteria (rugarch package output):

| Criterion      | Value    |
|----------------|----------|
| Log-Likelihood | -3603.79 |
| AIC            | 4.9435   |
| BIC            | 4.9617   |

## 4.3 Model Comparison

To evaluate the improvement of the GARCH(1,1) model over the constant variance benchmark, we compare their Information Criteria and Log-likelihood values.

| Model             | AIC    | BIC    | Log-Likelihood |
|-------------------|--------|--------|----------------|
| GARCH(1,1)        | 4.9435 | 4.9617 | -3603.79       |
| Constant Variance | 5.1907 | 5.1979 | -3787.18       |

The constant variance model assumes homoscedasticity, while the GARCH(1,1) model allows for time-varying conditional variance, which is better for clustering the volatility of Bitcoin.

It can be seen that for the GARCH(1,1) model, the Log-Likelihood is -3603.79, the AIC is 4.9435, and the BIC is 4.9617. For the constant variable model, using the descriptive statistics in Section 2.3, the Log-Likelihood is -3787.18, the AIC is 5.1907, and the BIC is 5.1979. Therefore, both AIC and BIC are lower for the GARCH(1,1) model than for the constant variance model. It indicates superior fit to the sample even after adjusting for additional parameters. So, the GARCH(1,1) model is preferred because it better captures the clustering of the volatility of Bitcoin's returns.

#### 4.4 Diagnostic Checks

##### Ljung-Box Test on Standardized Residuals

| Lag | Q-statistic | p-value |
|-----|-------------|---------|
| 5   | 7.06        | 0.2160  |
| 10  | 11.62       | 0.3116  |
| 20  | 15.64       | 0.7384  |

##### Ljung-Box Test on Squared Standardized Residuals

| Lag | Q-statistic | p-value |
|-----|-------------|---------|
| 5   | 2.23        | 0.8170  |
| 10  | 3.70        | 0.9598  |
| 20  | 11.35       | 0.9366  |

For the standardized residuals, all p-values are greater than 0.05, which means that there is no remaining linear dependence of the mean equation. It suggests that the mean of the GARCH(1,1) model is adequately specified. For the squared standardized residuals, all p-values are also much greater than 0.05 (no significant autocorrelation). Therefore, it indicates that the GARCH(1,1) model removes the ARCH effects observed earlier.

## 4.5 Interpretation

The empirical analysis results confirm that the clustering of volatility occurs in Bitcoin returns. The GARCH(1,1) model shows that Bitcoin volatility is highly persistent ( $\beta = 0.9462$ ,  $p < 0.001$ ), which means that high volatility will continue. Recent shocks have a small but important impact ( $\alpha = 0.0528$ ,  $p\text{-value} = 0.0292$ ). Also, the sum of the coefficients ( $\alpha + \beta = 0.9990$ ) is close to unity, meaning that shocks decay very slowly and periods of high volatility will persist.

From an asset management perspective, it means that traditional risk models underestimate probabilities of the extreme loss because Bitcoin has persistent volatility and fat-tailed distribution (shape = 3.33,  $p < 0.001$ ). In other words, the constant-variance risk model is inadequate for Bitcoin. Therefore, dynamic risk measures like Value-at-Risk (VaR) or Expected Shortfall are useful to quantify the tail risk (Li & Chen, 2025; Panayiotis & Stengos, 2021).

## 5. Conclusion

In this paper, the presence of volatility clustering was examined for Bitcoin daily returns in the period between January 2021 and December 2024 via a GARCH(1,1) methodology. As regards the empirical findings, there is evidence that there is indeed volatility clustering because of the powerful rejections of the null hypothesis of no ARCH effects based on the Ljung-Box test (the p-values being below 0.001 for lags 5, 10, and 20 in squared returns). The volatility clustering was estimated to be highly persistent by the GARCH(1,1) model ( $\alpha + \beta = 0.9990$ ), as well as the fact that the fat-tailed nature of Bitcoin return distributions could be well represented by the Student-t distribution. Finally, according to the AIC/BIC criteria, the GARCH(1,1) performed the best among different models under study.

Thus, there are several key takeaways regarding applications of the research findings for asset management practices. First, in view of persistent volatility and fat tails, risk measurement techniques based on constant variances would result in severe underestimations of risks, which means that GARCH-based VaR and ES calculations play an essential role in risk management.

This study comes with its own limitations. For instance, the model is constructed using only one cryptocurrency while the period covered by the data is just four years. Future studies may look into applying other cryptocurrencies or longer periods or even the application of the multivariate GARCH model in testing the spill-over effect. However, it is clear that time-varying volatility models should be applied in examining Bitcoin and other volatile currencies.

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